BM4 Implicit Method

Physical state and one reduced Hairer projection per complete cycle

Explicit palindromic operations

All ABBA and BM4 presets inherit `PalindromicMethod`. For signed $d = ch$, the elementary operations are

$$A_d(u, v, t) = (u + df(t, v), v, t), \qquad B_d(u, v, t) = (u, v + df(t, u), t), \qquad T_d(u, v, t) = (u, v, t + d).$$

Only T advances time. A coefficient $c = 1/2$ means duration $h/2$; there is no additional half inside A or B. The optional exact C flow rotates the copy difference and requires an independent coupling object. Both BM4 presets omit C.

In chronological order, standard ABBA2 is $A_{h/2}, B_{h/2}, T_h, B_{h/2}, A_{h/2}$. The distinct symmetric variant $T_{h/2}, A_{h/2}, B_{h/2}, B_{h/2}, A_{h/2}, T_{h/2}$ evaluates every spatial shear at the midpoint time. Both exhibit second order on the tested nonautonomous rotation. Arbitrary palindromes do not inherit fourth or sixth order, and an arithmetic projection need not be reversible or symplectic.

`Composition` denotes an explicit tuple of operations. Consecutive time translations combine as $T_a T_b = T_{a+b}$. The generic `PalindromicImplicit` and `PalindromicMidpoint` methods select one exterior Hairer projection or a final arithmetic mean. The signed spatial flows and passive momentum normalization are retained; observers reconstruct their groups from elementary traces without replay. NumPy and JAX execute the same explicit operations and differentiate their retained shear inputs. See `docs/models/palindromic/simulation/palindromic-operations.md`.

Current uncoupled configuration

Both `BM4Implicit` and `BM4Midpoint` omit the harmonic coupling flow and no longer accept a coupling-frequency constructor argument. Their twelve coefficients, signed clock and projection placement are unchanged. The independent operation `HarmonicCoupling` retains C for future compositions. Existing observer/archive rates are fixed at zero for new runs; nonzero historical calculations require their original frozen implementation.

1 Twelve-stage BM4 base map

Let χ_h be a first-order map and $\chi_h^* = \chi_{-h}^{-1}$ its adjoint. One complete step uses the palindromic composition

$$\Psi_h = \chi_{b_{12}h} \circ \chi_{b_{11}h}^* \circ \cdots \circ \chi_{b_2 h} \circ \chi_{b_1 h}^*,$$

with alternating adjoint and direct maps in chronological order and

$$(b_1, \ldots, b_{12}) = (a_1, a_2, a_3, a_4, a_5, a_6, a_6, a_5, a_4, a_3, a_2, a_1),$$

$$\begin{aligned} a_1 &= 0.0792036964311957, & a_2 &= 0.1303114101821663, \\ a_3 &= 0.2228614958676077, & a_4 &= -0.3667132690474257, \\ a_5 &= 0.3246481886897062, & a_6 &= 0.1096884778767498. \end{aligned}$$

The half-cycle coefficients sum to 1/2. Palindromy makes the composition self-adjoint and the order conditions give designed global order four. The negative coefficient a_4 is an intentional backward subflow.

2 Physical state and internal workspace

Twelve direct/adjoint stages enclosed by one spatial Hairer projection. Energy tracking is optional and passive. After convergence, passive energy quadrature evaluates $-\partial_t H$ at the 24 shear inputs retained from the accepted residual, with their original signed durations. It repeats no spatial map or coupling and performs no new nonlinear solve. Trial traces are discarded; energy evaluations remain outside physical solver counters. The common state conventions are specified below.

3 One Hairer projection around the complete cycle

For every complete BM4 step, the method displaces the diagonal embedding by a single multiplier $\mu \in \mathbb{R}^{2N}$,

$$\widehat{Y}_n = Ez_n + N\mu, \qquad M(\mu) = \Psi_{h,t_n}(\widehat{Y}_n), \tag{1}$$

$$r(\mu) = GM(\mu) + 2\mu, \qquad z_{n+1} = P\big(M(\mu) + N\mu\big). \tag{2}$$

The accepted multiplier is the root $r(\mu) = 0$. Thus the projection is applied once around the full twelve-stage composition, never between its internal stages.

With $J_{\mathrm{BM4}} = D\Psi_{h,t_n}(\widehat{Y}_n)$, Newton uses

$$D_\mu r = GJ_{\mathrm{BM4}}N + 2I.$$

The implementation forms J_{BM4} as the exact ordered product of the twelve guiding-centre stage Jacobians, with a centred-difference fallback. Good Broyden is retained as a nonlinear-solver choice; it changes the finite iteration path but not the defining projected map.

4 Fixed-grid lifecycle and diagnostics

The fixed controller advances accepted diagonal states on an output-independent grid. Off-grid samples use independent shadow steps. Only main steps contribute observations and nonlinear diagnostics: corrections, residual evaluations, residual norm, tolerance and multiplier norm. Coupling and Jacobian settings are recorded as metadata. Optional observers retain physical endpoints and twelve base-stage snapshots, allowing an independent energy reconstruction.

5 Implementation correspondence

The public class lives in `src/methods/palindromic/bm4_implicit.py`.

It directly inherits `ProjectedPalindromicMethod`, alongside ABBA2Implicit and ABBA4Implicit. That base owns preparation, solver validation and advancement; the concrete classes retain their recipe tuples, controls and observation adapters. Its twelve-stage recipe is defined in `bm4_midpoint.py`, and its reduced equation uses `projection.py`. See `implicit-reduced.tex` for the projection derivation. The former BM4 import adapters have been removed.

The public names and exports from `simulation` remain unchanged. Both BM4 constructors now omit harmonic coupling. Internal imports use the defining modules; old import adapters have been removed. The ordinary implicit path is `advance`, `solve_projection`, `compose`, then explicit $A/B/T/C$ operations. Arithmetic midpoint variants use one explicit average instead of a nonlinear projection. Adjoint stages evaluate at the start of their signed substep; direct stages evaluate at its end.

Newton and Broyden share the spatial equations and convergence services. Exact particle-block tangents, passive energy and optional observation use retained shear inputs; energy is evaluated

only after accepting the spatial map, with the existing factor 1/2 for physical κ. Time and momentum never enter the spatial root. The new internal family name does not enable the retired full time/momentum projection.

Current module responsibilities and diagrams are documented in `docs/models/palindromic/simulation/palindromic-simulation-architecture.md`.

6 Physical state and endpoint energy observations

Accepted states contain one physical vector $Z \in \mathbb{R}^{dN}$ and scalar time. GC vectors have $2N$ entries and FC vectors have $4N$; doubled workspaces are transient. InitialConditions supplies values and layout during preparation only. The numerical base retains functions, metadata, run status and copy count; execution owns the initial snapshot and output schedule. Supplied Jacobians are independent particle blocks (N, d, d); finite differences use dense matrices.

The generic `EnergyObserver` evaluates only

$$H_n = H(t_n, Z_n), \quad H_{n+1} = H(t_{n+1}, Z_{n+1}), \quad (\partial_t H)_n, \quad (\partial_t H)_{n+1}, \quad \Delta H_n = H_{n+1} - H_n.$$

It also evaluates these physical functions at requested saved samples, without consuming internal stages or integrating a momentum coordinate. The derivative is positive, analytic and partial at fixed state. GC differentiates its normalized Hamiltonian potential with `dt=1`; spatial splines and Fourier time factors retain their original scaling. This is not the secant $\Delta H_n/h$. Variation of a time-dependent Hamiltonian is not a numerical conservation error.

Specialized historical generalized-energy studies separately reconstruct $\dot{\kappa} = -\partial_t H$ from retained numerical stages. Their signed weights, physical normalization and archived diagnostics keep their original meaning. They are not part of generic energy observation. Host and JAX execution share the fixed numerical map and physical-only accepted state. JAX evaluates endpoint energy on the device without a momentum accumulator.

7 Optional JAX execution

An optional JAX driver executes the same numerical map on one selected CPU or GPU, with particle operations batched inside a compiled sequential time loop. It reuses the canonical coefficients and prepared potential splines. Physical-state tolerance scaling, signed stage times, passive energy weights and independent shortened output maps retain their definitions above. Applicable Newton systems use independent particle blocks; an explicitly selected Broyden iteration retains its global secant matrix. Nonlinear iteration limits and failure checks remain active on the device.

Both execution routes use double precision and return the same physical solution layout. Floating-point reassociation can produce round-off differences without changing the designed order or geometric properties. First-call compilation is separate from warmed integration timings, which include transfers and output construction. The sibling runtime document and shared execution contract specify supported callbacks and exact device scope.